



Lighting Installation Guide

Version History

Version	Revised By	Changes	Date
1.0	Mark Veikos	Initial Version	4/12/2018
1.1	Mark Veikos	Ceiling Mounted Restroom Lights, Freezer & Cooler Lights	8/1/2018
1.2	Lee Kaufman	General Changes	10/5/2018
1.3	Raphaelle Canaan	Added SubEnergy Branding	7/21/21
1.4	Raphaelle Canaan	Ceiling Mounted Fixtures, Counter Menu Lights, Occupancy Sensors, Ballast Warning & Romex Wiring	4/13/22
2.0	Austin Gattis	Complete revision, removed Subway specific brand standards	8/19/25

Last Saved: 8/19/2025 by Austin Gattis

1.Overview

This guide will detail the steps to replace all existing lighting fixtures with LED equivalents, as noted within the scoping document. Only the lights notated within the excel file as being scoped shall be installed!

Several fixture types, such as 2x2 and 2x4 fixtures, will be replaced with both color and wattage adjustable LED fixtures. These fixtures will be “tuned” during the installation to the specific conditions of the store. These standards are noted within the scoping file as well, though following the directions of the store manager is acceptable as well.

During the installation process, please consult with store manager if there are any questions related to lighting levels or if what is installed is acceptable.

Please note that for some sites, existing LED fixtures shall be replaced with newer, more energy efficient LED fixtures. This shall be noted on the scoping document, should this occur.

2. Install and Adjust Standard LED Fixtures per Scoping Doc

a. Install Fixtures Room by Room

All existing fixtures are documented within the scoping document, as well as their properties and their location. The scoping document covers which fixtures shall be replaced, which shall be relamped, which shall remain, as well as the brand standards of the site in question. It is important to follow the scoping document throughout the entire site, and if any deviations from it are necessary, these must be confirmed with Budderfly prior to installation.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Fixture ID	Ceiling Type	Room Notes	Fixture Height Greater than 14 ft	Room Type	Existing Fixture	Existing Fixture Description	Existing Fixture Qty	Existing Bulb Qty per Fixture	Existing Total Bulb Wattage	Ballast Factor	Existing Fixture Wattage (W)	Aggregate Existing Fixture Wattage	Usage (hrs/yr)	Existing Annual Aggregate
0	2 x 4 Tile		No	Back of House	Flat Panel	Flat Panel 2x4-LED-5000K-	23	1	32	1	32	736	8760	6447
1	Hard		No	Cooler/Freezer	Other	VT-VT-Fluorescent-T12-HO-5000K-	2	2	120	0.88	106	211	6000	1267
2	Hard		No	Cooler/Freezer	Jelly Jar	Jelly Jar-JJ-LED-A19-E26(medium standard screw base)-4000K-	1	1	15	1	15	15	6000	90
3	Hard		No	Cooler/Freezer	Other	VT-VT-Fluorescent-T12-HO-4000K-	2	2	120	0.88	106	211	6000	1267
4	Hard		No	Cooler/Freezer	Jelly Jar	Jelly Jar-JJ-LED-A19-E26(medium standard screw base)-4000K-	1	1	15	1	15	15	6000	90
5	2 x 4 Tile		No	Office	Flat Panel	Flat Panel 2x4-LED-5000K-	1	1	32	1	32	32	6000	192
6	2 x 2 Tile		No	Break	Recessed Can	Recessed Can-6" dia.-LED-5000K-	2	1	12.5	1	13	25	6000	150
7	2 x 2 Tile		No	Front of House	Recessed Can	Recessed Can-6" dia.-LED-4000K-	30	1	12.5	1	13	375	6000	2250
8	2 x 2 Tile		No	Front of House	Recessed Can	Recessed Can-6" dia.-LED-4000K-	12	1	13.5	1	14	162	6000	972
9	Hard		No	Front of House	Pendant	Pendant-Deco-LED-2700K-	2	1	40	1	40	80	6000	480
10	2 x 2 Tile		No	Front of House	Pendant	Pendant-Deco-LED-G40-E26(medium standard screw base)-2700K-	2	1	6	1	6	12	6000	72
11	Hard		No	Restroom	Recessed Can	Recessed Can-6" dia.-LED-4000K-	7	1	12.5	1	13	88	6000	525
12	Hard		No	Restroom	Recessed Can	Recessed Can-6" dia.-LED-4000K-	7	1	12.5	1	13	88	6000	525
13	Hard		Yes	Exterior	Pole	Pole-Pole-LED-5000K-	8	1	75	1	75	600	6000	3600
14	Hard		No	Exterior	Recessed Can	Recessed Can-Recessed-LED-5000K-	14	1	20	1	20	280	6000	1680

G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Existing Fixture Description	Existing Fixture Qty	Existing Bulb Qty per Fixture	Existing Total Bulb Wattage	Ballast Factor	Existing Fixture Wattage (W)	Aggregate Existing Fixture Wattage	Usage (hrs/yr)	Existing Annual Aggregate e Pow	Existing EM Ballast? (Y/N)	Existing Qty of Emergency Fixtures	Return Air Fixture? (Y/N)	Existing Qty of Non-Operational Fixtures	Does the Fixture Pass NEC? (Y/N)	Switching / Control Type	Qty Occ. Sensor - Proposed	Proposed LED Product
Flat Panel 2x4-LED-5000K-	23	1	32	1	32	736	8760	6447	No	0	No	0	Yes	None		N/A - Existing to Remain
VT-VT-Fluorescent-T12-HO-5000K-	2	2	120	0.88	106	211	6000	1267	No	0	No	0	Yes	None		TUBE 4000-5000-6500K 18W 120-23N
Jelly Jar-JJ-LED-A19-E26(medium standard screw base)-4000K-	1	1	15	1	15	15	6000	90	No	0	No	1	Yes	None		N/A - Existing to Remain
VT-VT-Fluorescent-T12-HO-4000K-	2	2	120	0.88	106	211	6000	1267	No	0	No	0	Yes	None		TUBE 4000-5000-6500K 18W 120-23N
Jelly Jar-JJ-LED-A19-E26(medium standard screw base)-4000K-	1	1	15	1	15	15	6000	90	No	0	No	1	Yes	None		N/A - Existing to Remain
Flat Panel 2x4-LED-5000K-	1	1	32	1	32	32	6000	192	No	0	No	0	Yes	None		N/A - Existing to Remain
Recessed Can-6" dia.-LED-5000K-	2	1	12.5	1	13	25	6000	150	No	0	No	0	Yes	None		N/A - Existing to Remain
Recessed Can-6" dia.-LED-4000K-	30	1	12.5	1	13	375	6000	2250	No	0	No	0	Yes	None		N/A - Existing to Remain
Recessed Can-6" dia.-LED-4000K-	12	1	13.5	1	14	162	6000	972	No	0	No	0	Yes	None		N/A - Existing to Remain
Pendant-Deco-LED-2700K-	2	1	40	1	40	80	6000	480	No	0	No	0	Yes	None		N/A - Existing to Remain
Pendant-Deco-LED-G40-E26(medium standard screw base)-2700K-	2	1	6	1	6	12	6000	72	No	0	No	0	Yes	None		N/A - Existing to Remain
Recessed Can-6" dia.-LED-5000K-	7	1	12.5	1	13	88	6000	525	No	0	No	0	Yes	Occupancy Sensor		N/A - Existing to Remain
Recessed Can-6" dia.-LED-4000K-	7	1	12.5	1	13	88	6000	525	No	0	No	0	Yes	Occupancy Sensor		N/A - Existing to Remain
Pole-Pole-LED-5000K-	8	1	75	1	75	600	6000	3600	No	0	No	0	Yes	None		N/A - Existing to Remain
Recessed Can-Recessed-LED-5000K-	14	1	20	1	20	280	6000	1680	No	0	No	0	Yes	None		N/A - Existing to Remain

Site Notes:

- New Flat Panels are only to be installed into drop ceilings. If your site calls for install of Flat Panels into Hard ceiling, Coordinate with Budderfly Project Manager.
- Interior, Front of House Brand Standards: 3500K
- Interior, Back of House Brand Standards: 4000K
- Exterior Brand Standards: 5000K
- Special Brand Standards: N/A

b. 2x4 or 2x2 Recessed Flat Panels

All existing non-LED troffers shall be replaced with new integral LED flat panels, when they are located in a recessed grid ceiling. Those that are located within a hard ceiling shall be relamped. Flat panels shall never be installed in a hard ceiling.

When installing integral LED 2x4s or 2x2s, please ensure the wattage and color temperature settings are set to those within the scoping document. The default wattage for all 2x4s and 2x2s should be 23w.

In some locations, existing LED flat panels will be replaced with newer, more efficient LED panels. Please consult the scoping document to confirm this approach.



c. Ceiling Mounted Fixtures

Ceiling mounted fixtures will remain in place and should not be removed from the ceiling.

These lights typically appear in restrooms and back areas. Instead of replacing the fixture, bulbs will be re-lamped with LED tubes. Existing bulbs are typically 4ft T8 tubes.



d. Emergency Lighting Bugeyes / Exit Signage



In the event of a non-LED exit sign or non-LED exit sign with emergency lighting combo, replace with LED Exit Sign with or without emergency lights as specified within the scoping doc.

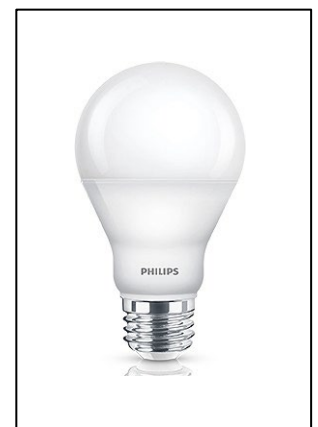
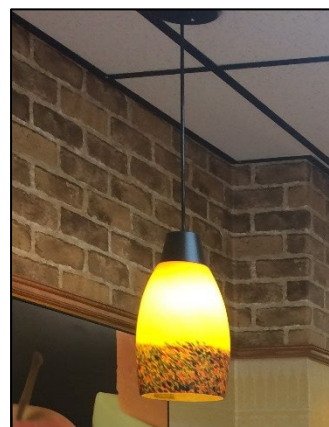
Existing bugeye fixtures, even those that are non-LED, will typically be left existing. Please refer to the scoping document to confirm.

e. Emergency Backup Lights

Appropriately address emergency backup lighting, ensuring compliance with all regional codes and safety standards. Emergency backups are typically found within existing 2x4s, 2x2s, and in some can lights. Can light batteries are typically remote, while the 2x4/2x2s are typically integrated within the existing fixture. When replacing a fixture with an emergency backup, never replace it with a standard fixture! The new fixture must also have an emergency backup as well. If one is missing, please consult Budderfly for further direction.

f. Pendant Lights

Pendant lights hang down from the ceiling and contain screw-in A19 bulbs or other types of screw-in bulbs. Currently, the pendants may have either incandescent, compact fluorescent, or LED bulbs. If the bulbs are all incandescent or CFLs, replace entirely with LEDs, as per the scoping doc.



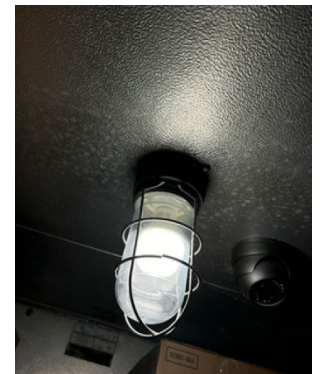
g. Can Lights

Some sites may have recessed can lights throughout. These may be PAR lamps, PL's, CFLs, MR16's, integral LED, or others. Can lights in a hard ceiling will be relamped, will those in a grid ceiling will typically be replaced entirely. Please refer to the scoping document to confirm, and please reach out to Budderfly if there are issues with the proposed installation.



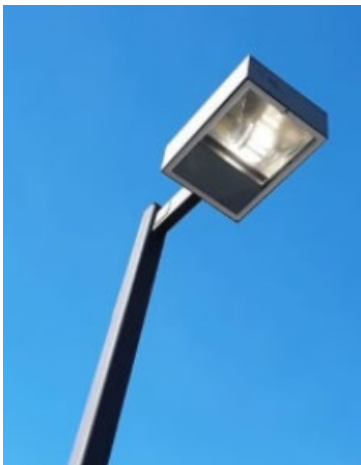
h. Walk-in Freezer and Cooler Lights

Walk-in freezer and cooler typically contain “jelly jar” or surface mounted strip fixtures. These fixtures will typically be relamped if non-LED, while existing LED fixtures will remain. Please consult the scoping doc to confirm the exact approach.



i. Exterior Lights

In locations where the store is responsible for exterior lighting (not in strip malls or combined buildings), exterior non-LEDs shall be replaced with LEDs, as noted withing the scoping document. Multiple types of exterior lights may be found at each site, including wallpacks and floodlights, downlights, and parking lot lights. Replace these with LED bulbs / fixtures of the equivalent wattage, making sure *only* those that are owned by the client are replaced.



3. Occupancy Sensors

Occupancy sensors should be installed in only “small rooms” (restrooms, office, utility closets, etc.), as specified within the scoping document. Occupancy sensors should not be installed in bathrooms where partitions (toilet or urinal) are present as they could prevent the motion sensor from activating during movement.



For the occupancy sensor to function, a neutral conductor must be present at the switch. If the current light switch does not have a neutral, please consult with the Budderfly project manager.

4. Post-Install

We will require you to fill out a post-install survey once a project is complete. This must be completed within 24-48 hours from completion of installation. Failure to do so will result in rejected invoicing. If a return trip is needed, then a post install survey will be done at the end of the initial trip and downloaded into our system.

5. Code Compliance Issues

It is the responsibility of the contractor to ensure that they are code compliant when performing work on these sites. Any NEC or local code wiring violations discovered during the DIRECT installation of the lights (improper wiring, exposed junction boxes, exposed bus bars at lighting panels, etc) should be corrected as part of the lighting installation. All code violations, including those not related to the direct installation of the lighting scope of work, shall be reported to the Budderfly PM, and if it is related to the direct installation, a quote necessary to fix the violation will need to be submitted to the PM for review as well.

Some common concerns include:

- Romex wiring, which in general is not permitted in commercial applications, though may be allowed to be grandfathered in within certain jurisdictions. It is up to the local Electrician to use his or her knowledge and advise if the existing Romex wiring may be re-used. Under no circumstances should the lighting fixtures be replaced if Romex wiring is not permitted by the authority having jurisdiction (AHJ).
 - Improper wiring
 - Exposed junction boxes
 - Exposed bus bars at lighting panels
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6. Go-Backs

It is often necessary for a site to have a return visit. If it is necessary, the following rules dictate the cost responsibilities of the go-back trip:

- If the cause of an installation go back is due to an inaccurate survey, contractors will be responsible for the labor to complete a return trip.
- If the cause of an installation go back is due to an installation quality issue, the contractor will be responsible for the labor to complete a return trip.
- If the cause of an installation go back is due to Budderfly inaccuracy, we will be responsible for the labor to complete a return trip.

7. Misc.

- Follow all local codes during installation. This includes earthquake certification and stickers when fixtures or bulbs are powered by line voltage.
- Please ensure adequate protection for customers, workers, and food products during the installation process.
- Please ensure proper PPE (Personal Protective Equipment) is used throughout the entire duration of the install.
- All installs must follow OSHA and NFPA 70E Rules and Regulations throughout.
- We require proper disposal of all existing fixtures and lamps. Please retain all supporting documentation.
- Please ensure that if a lift is needed that this is acquired in advance and approved by both Budderfly and the client. Please follow all necessary safety precautions when operating a lift.